Neural Computing and Applications (2022) 34:4781–4801
https://doi.org/10.1007/s00521-021-06667-3

ORIGINAL ARTICLE

Deep neural network-based relation extraction: an overview

Hailin Wang[1] · Ke Qin[1] · Rufai Yusuf Zakari[1] · Guoming Lu[1] · Jin Yin[1]

Received: 8 February 2021 / Accepted: 27 October 2021 / Published online: 5 January 2022

Abstract

Knowledge is a formal way of understanding the world, providing human-level cognition and intelligence for the next-generation artificial intelligence (AI). An effective way to automatically acquire this important knowledge, called Relation Extraction (RE), plays a vital role in Natural Language Processing (NLP). To date, there are amount of studies for RE in previous works, among which these technologies based on deep neural networks (DNNs) have become the mainstream direction of this research. In particular, the supervised and distant supervision methods based on DNNs are the most popular and reliable solutions for RE, whose various evolutions on structure and settings have affected this task. Understanding the model structure and related settings will give the researchers a deep insight into RE. However, little research has been done on them. Hence, this paper starts from these two points and carries out analysis around the mainstream research routes, supervised and distant supervision. Meanwhile, we classify all related works according to the evolution of model structure to facilitate the analysis. Finally, we discuss some challenges of RE and give out our conclusion.

Keywords Overview · Information extraction · Relation extraction · Neural networks

1 Introduction

Artificial intelligence (AI) integrating knowledge is a hot topic in current research. It provides human thinking for AI to solve complex tasks. One of the most important techniques for supporting this research is knowledge acquisition, also called relation extraction (RE). One aim of RE is to process human language text, to find unknown relational facts from a plain text, organizing unstructured information into structured information. For example, there is a sentence: “< e1> Jobs </e1> is the founder of < e2> Apple</e2>.” It marks the two entities “Jobs” and “Apple” by a pair of XML tags. From the sentence, the RE model outputs a triple (Jobs, Apple, founded_by). With the amount of these triples, a knowledge base (KB) could be constructed, which is useful for many downstream applications, such as knowledge completion, question&answering, text generation, and summarization. These downstream tasks also further advance AI.

However, constructing a large-scale KB remains a difficult task. Although existing KBs including Freebase [1], DBpedia [2], Wikidata [3], and YAGO [4] have billions of triples, they are still far away from being perfect. In general, the above KBs are constructed by manual, being time-consuming and laborious. As an increasing amount of data are made available online and millions of unstructured texts drastically increase every day, the manual methods are no longer suitable for those texts. Recently, RE task performs an automatic way to distilled relational facts, which could deal with this situation without human intervention. At present stage, mature RE methods include hand-built pattern methods, semi-supervised methods, supervised methods, unsupervised methods, and distant

✉ Ke Qin
qinke@uestc.edu.cn

Hailin Wang
lynn_whl@msn.com

Rufai Yusuf Zakari
rufaig6@gmail.com

Guoming Lu
lugm@uestc.edu.cn

Jin Yin
yinjin@wchscu.cn

[1] Trusted Cloud Computing and Big Data Key Laboratory of Sichuan Province, School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China

supervision methods. Their development goes from DIPRE (Dual Iterative Pattern Relation Expansion) [5], Snowball [6] to DNN-based methods including the model first using CNN on RE [7], the attention mechanism with BiLSTM [8], and different neural structures with shortest dependency path [9]; especially, some supervised and distant supervision methods based on DNNs have achieved superior performance.

Hence, RE has attracted extensive attention and promoted its application in industrial fields, including biology [10], medicine [11], finance [12], and so on, which in turn have driven the technology's progress. Furthermore, a lot of researchers report RE. Kumar [13] focused on early works about supervised and distant supervision methods. Pawar et al. [14] surveyed several mainstream technology fields from the perspective of traditional methods. Smirnova and P Cudré-Mauroux [15] reviewed distant supervision methods, the embedding-based approaches of which just gave a summary analysis of convolutional neural network methods. Nayak et al. [16] covered many aspects of RE, including zero-shot, distant supervision, and joint RE, while this work's analysis about distant supervision just considered the de-noise methods. Other works [17–19] also provided new sight into this field, concentrating on performance, promising directions, and hot topic, respectively.

With the increase in various research studies, RE models based on DNNs have also undergone various evolutions along with the continuous development of DNNs. And different model structures and other settings have different influences on RE task, but we found that how various DNNs structures promote the RE task is still a gap, needing further analyzing. As far as we know, rarely previous research has investigated them.

Consequently, this article considers the field of analyzing model structures and other settings (including input feature set, loss function, etc) as the main subject of its study. We present an extensive survey and give a comprehensive introduction of RE to the prevalent DNN-based methods. To begin with, Section 2 introduces the premise of RE frameworks, including a general framework and some basic conceptions of RE. Section 3 lists the traditional methods of RE and discusses some obstacles of them. In Section 4 and 5, various DNN-based methods will be classified into different types to compare in detail, considering the structure and different settings. Section 7 further provides an analysis of some problems or future research directions. To the best of our knowledge, this is the first survey paper that compares so many different model structures and settings of RE in detail.

2 Premise

2.1 General framework

By conducting an extensive literature review, we summarized the framework of DNN-based RE methods as four components, which are shown in Fig. 1. The detailed description of these four components is as follows:

Data sets: The supervised data sets (SemEval 2010-task8 [20] and FewRel [21, 22]) are often obtained by manual annotation with high accuracy and low noise, but small size. Instead, distant supervision data sets usually are acquired by a physical alignment of entities between a corpus and KB, which have a bigger size and high multi-domain applicability (e.g., Riedel et al. [23]), but low accuracy and high noise. In a sense, this component is not a simple module in general DNN-based models. But in the RE field, especially in distant supervision, the physical align phrase, constructing training triples, plays a vital role in RE.

Sentence Representation: In NLP field, to understand human language for computers, words are usually represented as a series of real value vectors, such as word2vec and Glove methods. Meanwhile, the position embedding is introduced to better express the positional relation between words and the entity pair. Hence, the final representation of the words is the combination of word vector and position embedding. Consequently, the final sentence representation is composed of these word representations.

Feature extraction: In general, these DNN-based methods are fed with the sentence representation above. With the annotated data sets, these methods output a feature extractor by training. And this extractor can extract high-level features from the above sentence representation.

Classifier: With the high-level features and a predefined relation inventory, the classifier outputs the relation between the entity pair in the sentence and then evaluates the result.

2.2 Basic conception

In addition to the above general frameworks, the basic concepts of DNN-based RE system commonly used in these frameworks are as follows:

Neural Networks have been widely used in image processing, language processing, and other fields [24] in recent years, with very remarkable results. Researchers have designed many kinds of DNNs, including Convolutional Neural Networks (CNNs) [25], Recurrent Neural Networks (RNNs) [26], Recursive Neural Networks [27], and Graph Neural Networks (GNNs) [28]. Different kinds of DNNs have different structures and advantages in

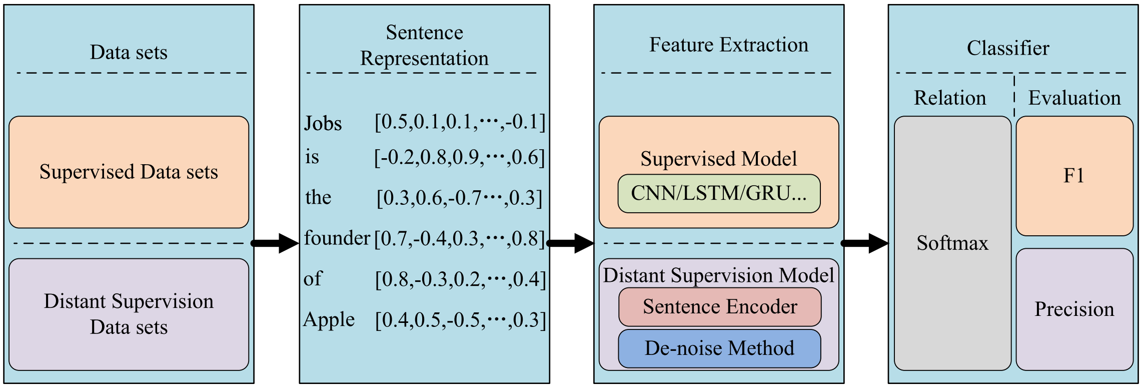

Fig. 1 General Framework of DNN-based RE

dealing with various language tasks. For example, CNNs, with parallel processing ability, are adept at processing local and structural information. Instead, the RNNs, having advantages in dealing with a long text, cope with time-series information by considering the factors before and after data input. Moreover, developing gradually in recent years, the GNNs, another kind of neural network, process data with a graphical structure; for example, the grammatical dependency parse tree, a general graphical structure for RE, is suitable for the GNNs. In addition to the above-mentioned commonly used networks, there are also some other RNNs variant networks used in RE systems, such as LSTM (Long Short Term Memory network) [29–31] and GRU (Gated Recurrent Unit) [32].

Word Embedding is a method used to represent words in NLP, leveraging uniform low-dimensional, continuous, real-valued vectors to represent language. One of the earlier forms is one hot, which exits some problems like data sparsity, no meaning, dimensional disasters. To solve these problems, some scholars [33] proposed a new method called word2vec to overcome these disadvantages. In this way, all the word vectors are distributed, the dimensions of vector can be arbitrary, generally in 50 to 100 dimensions, and the value of the element can be any real value. This method has two advantages: one is that semantic information and context information of words can be obtained; the other is that similarity of words can be calculated by simple addition and subtraction. Hence, word2vec is a common component in DNN-based RE. Aside from the word2vec method, some researchers also designed other methods [34, 35].

Position Embedding (PE) provides a uniform way for the RE model to be aware of word positions. In RE task, the CNN-based models lack judgment on the word location information. To address this issue, Zeng et al. [36] proposed the position feature (PF), which will be adopted in subsequent methods of using CNNs [37–39], RNNs [40–42], and mixed frameworks [43, 44]. The PF is a

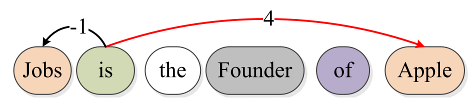

Fig. 2 Example of relative distance

combination of the relative distance between each word around the labeled entities in the sentence. For instance, given labeled entities: “Jobs” and “Apple”, in the sentence “Jobs is the founder of Apple”, the relative distance of the word “*is*” to “*Jobs*” is -1, to “*Apple*” is 4 (shown in Fig. 2). In this way, the distance of words around the entity words in the sentence can be expressed clearly. Furthermore, to make the model easy to understand the PF, the above two real values are mapped into a new vector space, namely the position embedding process. Normally the dimension of this vector is 5. One example is shown in Fig. 2.

In addition to PE, some RNN-based methods are also used <e> position indicators (PI) to further enhance the representation of entity pair. In SemEval 2010-task8, this data set uses four position indicators to indicate the entity pair in the sentence. The example is shown in Table 1.

Shortest Dependency Path (SDP) is a word-level de-noise method and derived from a grammatical dependency tree, which masks irrelevant words influencing the relation of entities in a sentence. (The grammatical dependency tree can be obtained by the Stanford Parser[1]). Bunescu and Mooney [45] first used the SDP to design a kernel-based method to finish RE task, and then, a lot of SDP-based models followed this work, such as SDP-LSTM [46], BRCNN [9], DesRC(BRCNN) [43], Att-RCNN [47], and FORESTFT-DDCNN [48]. For instance, a sentence “< e1> People</e1> have been moving back into < e2> downtown</e2>.” can be parsed to an SDP: “$[People]_{e_1}$ →moving→into→ $[downtown]_{e_2}$.” This example illustrates

[1] This tool in this work can be downloaded from here http://nlp.stanford.edu/software/lex-parser.shtml.

Table 1 Example sentence with position indicators

Example:	< e1> Jobs </e1> is the founder of < e2> Apple</e2>.
Indictors:	< e1>, </e1>, < e2>, </e2>

that the SDP captures the predicate-argument sequences. In a sense, these sequences have great benefits for RE. Firstly, this method compresses the information content of sentences. Secondly, it directly shows the dependency relations between each word. Finally, it also provides a clearer relation direction between entities.

3 Traditional methods

3.1 Categories of methods

The existing RE approaches can be divided into **hand-built pattern methods, semi-supervised methods, supervised methods, unsupervised methods**, and **distant supervision methods.** In this paper, we refer to the five methods without DNNs as traditional methods.

Hand-built pattern methods require the cooperation between domain experts and linguists to construct a knowledge set of patterns based on words, part-of-speeches (POSs), or semantics. With this linguistic knowledge and professional domain knowledge, RE can be realized by matching the preprocessed language fragment with the patterns. If they match, the statement can be said to have the relation of the corresponding pattern [49, 50]. Table 2 is an example of a pattern for hyponymy.

Semi-supervised methods are pattern-based methods in essence. The typical method is a bootstrapping algorithm, and the representative model is DIPRE proposed by Brin et al. [5]. The idea behind this method is first to find some seed tuples with high confidence; the bootstrapping algorithm extracts patterns with the tuples from a large number of unlabeled corpus. Then, these patterns can be used to extract new triples, the process of which is shown in Fig. 3. Some other representative models are: Snowball [6], KnowItAll [51], TextRunner [52]. There is also a recent method proposed by Phi et al. [53].

Unsupervised methods adopt a bottom-up information extraction strategy based on the assumption: the context information of different entity pairs with the same semantic relation is relatively similar. An earlier unsupervised approach was proposed by Hasegawa et al. [54]. This extraction process could be divided into three steps: step 1 to extract an entity pair and its context, step 2 to cluster the entity pair according to the context, and step 3 to annotate the semantic relation of each class or describing the relation type.

Supervised methods consider RE as a multi-class classification problem. These approaches are classified into two types, feature based and kernel based [14]. In the feature-based methods [55, 56], each relation instance in the labeled data is used to train a classifier fed with subsequent new instances for classification. Generally, these features come from useful information (including lexical, syntactical, semantic) extracted from an instance context. Without a proper feature selection, a feature-based method is difficult to improve the performance. Compared with the feature-based methods, the kernel-based [45, 57] methods need rarely explicit linguistic preprocessing steps. But it depends more on the performance of the kernel function designed. The key step in this approach becomes how to design an effective kernel.

Distant Supervision methods are a kind of knowledge-based or weakly supervised methods proposed by Mintz et al. [58]. All of these works are based on this assumption: If two entities participate in a relation, all sentences that mention these two entities can express that relation. In other words, any sentence that contains a pair of entities participating in a KB is likely to express that relation. In this way, distant supervision attempts to extract the relations between entities from the text by using a KB, such as Freebase, as the supervision source. When a sentence and KB refer to the same entity pair, this method marks the sentence heuristically with the corresponding relation in the KB. For example, "Jobs is the founder of Apple."; in this sentence, the person "Jobs" and the organization "Apple" appear in Freebase, and Freebase has a triple (entity1: Jobs, entity2: Apple, relation: *founded_by*) corresponding to the mentioned entity pair. Therefore, these entities express a relation *founded_by*. The process is shown in Fig. 4.

3.2 Discussion

Although there are numerous ways to solve the problem of RE, this field has consistently shown that these methods exit various obstacles. The hand-built pattern and semi-supervised methods required the manual exhaustion of all relation patterns, resulting in inevitable human errors. In the supervised method, a variety of mature NLP toolkits [59] provided technical support for these approaches, but both feature design and kernel design were still time-consuming and laborious. The clustering results generated by unsupervised methods were generally broad, and one of the main obstacles was how to define an appropriate relation inventory. Moreover, this method had limited processing capacity for a low-frequency entity pair and also lacked a standard evaluation corpus or even unified evaluation

Table 2 Example of pattern “such Y as X”

Pattern	such Y as X
Corpus	... works by such authors as Herrick, Goldsmith, and Shakespeare.
Relation	Hyponym (“author”, “Herrick”), Hyponym (“author”, “Goldsmith”), Hyponym (“author”, “Shakespeare”)

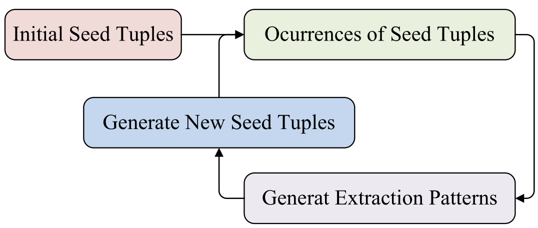

Fig. 3 Main idea of DIPRE

criteria. The distant supervision could effectively label data for RE, yet suffered from the wrong label and low accuracy problem. In addition, all of these approaches have domain limitations, error propagation, and poor ability to learn underlying features.

To solve these problems, some scholars try to adopt DNN-based methods to improve the performance of RE. In fact, in other fields of NLP, DNNs have been widely applied, such as machine translation, sentiment analysis, summarization, question&answering, and information recommendation, and all of them have achieved state-of-the-art performance. To date, DNN-based RE methods have been used in supervised and distant supervision RE mentioned above. These DNN-based methods [7, 36, 46] can automatically learn features instead of manually designed features based on various NLP toolkits. At the same time, most of them have completely surpassed the traditional methods in effect. Table 3 shows some comparison of traditional RE methods and earlier DNN-based methods, which illustrates that DNN-based methods can obtain higher scores with fewer features.

Based on the five traditional methods mentioned above, DNN-based methods introduced in this paper mainly focus on supervised methods and distant supervision methods, which are more suitable for practical application, and their structures and features set used in models are more presentative.

4 DNN-based Supervised RE

For the sake of simplicity and clarity, this paper further subdivides the methods using different DNNs as four types (e.g., **CNN, RNN or LSTM, Mix-structure**), and their evolutionary process is shown in Fig. 5a. Although further subdivision is carried out, the advantages of high accuracy of DNN-based supervised methods are consistent with the traditional supervised methods, and they have been further improved. The architecture of the supervised methods is shown in Fig. 5b. To facilitate the analysis, the development process of each type of model set is divided into two sub-types according to the evolution of model structure: **structure oriented** and **semantic oriented**. The structure-oriented classes improve the ability of feature extraction by changing the structure of the model; the semantic-oriented classes improve the ability of semantic representation by excavating the internal association of text.

Fig. 4 Process of distant supervision. The upper left of figure is the KB, and the lower left is the corpus source. After the text alignment process in the middle, the right side produces the packages corresponding to the KB to represent various relationships. Each package represents a relationship label, and these packages contain several sentence instances

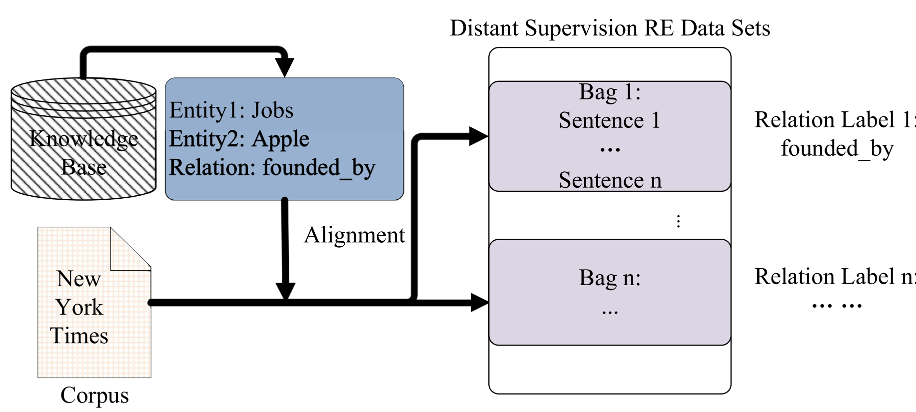

Table 3 Comparison of traditional methods and DNN-based methods

Classifier	Feature Sets	F1
SVM	POS, stemming, syntactic patterns	60.1
SVM	Word pair, words in between	72.5
SVM	POS, stemming, Syntactic patterns	74.8
MaxEnt	POS, morphological, noun compound, thesauri, Google n-grams, WordNet	77.6
SVM	POS, prefixes, morphological, WordNet, Dependency parse, Levin Classed, ProBank, FrameNet, NomLex-Plus, Google n-gram, paraphrases, TextRunner	82.2
RNN	POS, NER, WordNet	77.6
MVRNN	POS, NER, WordNet	82.4
CNN+softmax	Word pair, Words around word pair, WordNet	82.7

Some traditional classifiers, their feature sets and the F1-score for RE [36]. The F1 values are based on semeval2010-task8 [20]

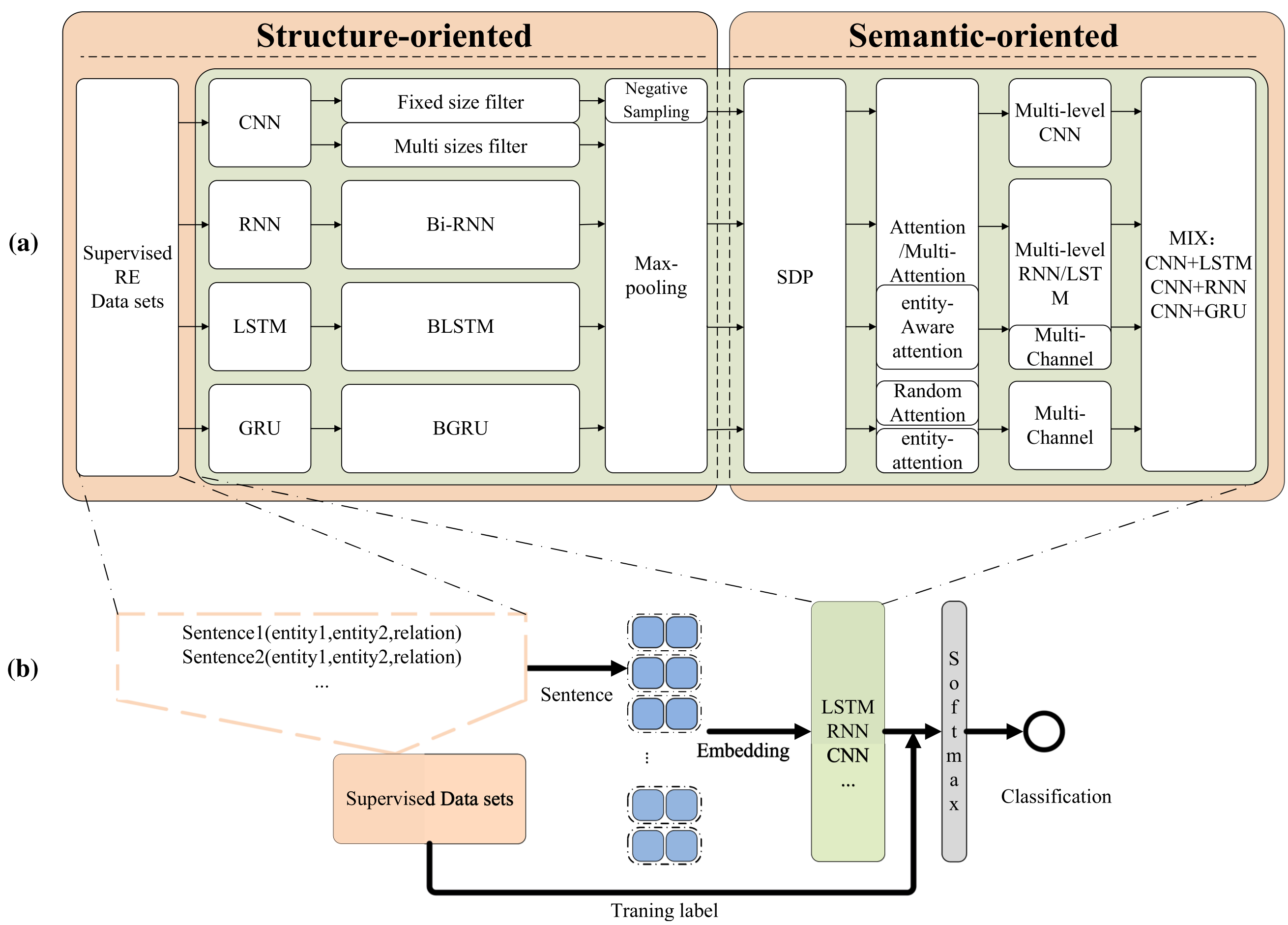

Fig. 5 **a** Is the evolutionary process, and **b** is the architecture of supervised methods. All of them are based on DNNs

4.1 CNN-based methods

CNN-based models are a general model of RE, which have achieved excellent results. In these models, a part of them play a key role in the following RE task, especially some modules, such as CNN with multi filters [37], piecewise convolution [61], attention mechanism, PE [36]. In the following, this subsection will introduce these related models. Some comparisons of model structure and settings are shown in Table 4.

Structure oriented: The first model using CNN on RE was proposed by Liu et al. [7]. With the synonym

Table 4 Comparison of CNN-based methods

Class	Author	Model name	Structure	Features set	Loss	Opt	F1
Str	Liu et al. [7]	–	Fixed size filter CNN	One hot	–	–	–
	Zeng et al. [36]	–	Fixed size filter CNN	WE1+WA+WN+PE	C	SGD	82.7
		–		WE1+WA+WN			69.7
	Nguyen et al. [37]	–	Multi sizes filter CNN+max-pooling	WE2+PE	–	SGD	82.8
	Santos et al. [38]	CR-CNN	Fixed size filter CNN+max-pooling	WE2+PE	R	SGD	84.1
		–			C		82.4
Sem	Xu et al. [60]	depLCNN+NS	Fixed size filter CNN+max-pooling+Negative Sampling	WE1+WA+WN+SDP	C	SGD	85.6
		–	Fixed size filter CNN+max-pooling	WE1+PE			83.7
	Wang et al. [39]	–	Fixed size filter CNN+two level CNN+two level attention+max-pooling	WE2+PE+WA	D	SGD	88
		Att-Pooling-CNN	Fixed size filter CNN+two level CNN+max-pooling	WE2+PE+WA			86.1

In this table, the Str and Sem refer to structure-oriented classes and semantic-oriented classes. The WE1, WE2, WE3 refer to Word embedding proposed by Turian et al. [34], Mikolov et al. [33], Pennington et al. [35], respectively. The WA, WN, PE, PI refer to Word around nominals, WordNet, Position embedding, Position indicator. The SDP, GR, WNSYN, Relative-DEP, Opt refer to shortest dependency path, grammar relation, hypernyms, relative dependency, Optimization. The C, D, R refer to cross entropy, distance function, ranking loss. All of the F1 values are based on semeval2010-task8 [20]. This table lists the best result of the model and its variants, and subsequent tables follow this format.

dictionary and any other lexical features, this model transformed the sentence into a series of word vectors, which was fed to a CNN and a softmax output layer to get a classification probability. The important contribution of this paper was to pioneer the application of DNNs in this task. Better result as it has achieved, this method still depended on many features from NLP toolkits, which barely considered the semantics, model structure, and feature selection.

To improve the feature selection, Kim [61] introduced multiple filters and max-pooling modules for CNN, which was probably one of the earliest works in the text classification task. Based on this work, Kalchbrenneret et al. [62] described a dynamic CNN that used a dynamic k max-pooling operator to pick up some features from the result of CNN. Both of the above two models [61, 62] achieved high performance on different tasks.

To finish this work with fewer NLP toolkits, Zeng et al. [36] creatively put forward the concept of PE between entities and combined the lexical information of entities with the sentence-level features extracted by CNN and integrated these features into the network, achieving state of the art on the task of SemEval-2010task8 in 2014. This model solved the cumbersome preprocessing problem in the task and avoided the error propagation problem to some extent. After this model, almost all CNN models have used the PE method and tried to extract information with fewer NLP toolkits, or even tried to extract relations without using any information other than word embedding. However, with a fixed size filter CNN, this model only focused on the local features and ignored the global features.

To bring more structure information, Nguyen et al. [37] combined the concept of multi window sizes filter for RE based on Kim's [61] work. Compared with fixed size filter CNN, this paper demonstrates that multi window sizes filter CNN brings more structured information to the model. This would be an efficient way to improve the CNN architecture. Many subsequent models have adopted this technique as well. Meanwhile, this paper also shows that the word vectors pre-trained and changing dynamically with model training are helpful to improve the performance. But in fact, dynamic word vector is not often referenced. The model structure is shown in Fig. 6.

Nevertheless, all the above models deploy the softmax module, which cannot eliminate the influence of other similar classes affecting the final classification results. Santos et al. [38] improved the loss function instead of the softmax classifier. The model's parameters were trained by minimizing a new ranking loss function (CR-CNN) over the training set, giving a higher probability to the correct class and lower probability for the wrong classes. This new loss function improved this model and has been used in other classifiers.

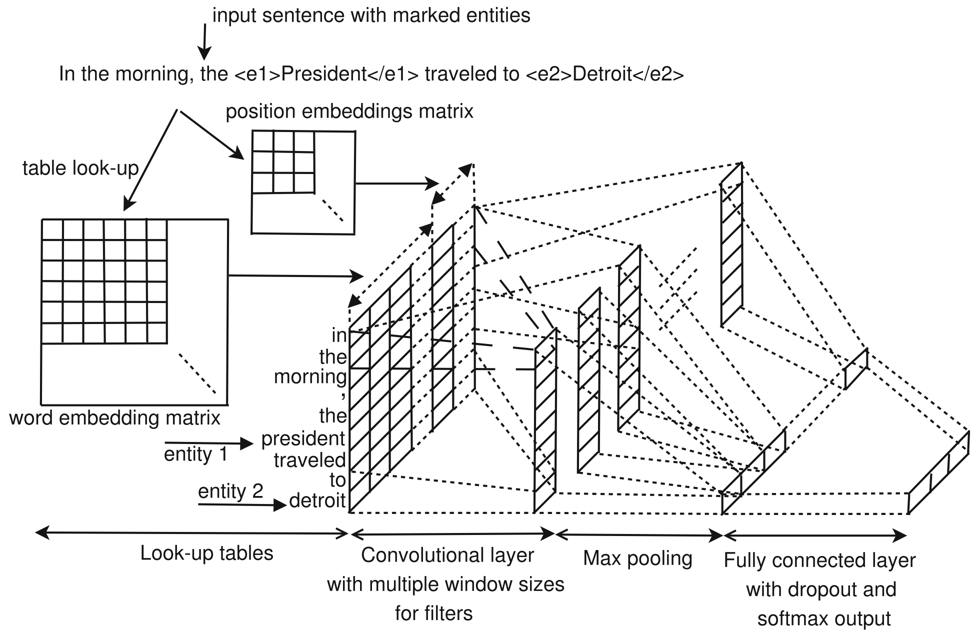

Fig. 6 Structure of model [37] with multi sizes filters

Semantic oriented: To learn more robust relation representations, Xu et al. [60] proposed a model through a CNN with the SDP. The model, taking the subject to object of a sentence as input, removes the words that are not related to the relation discrimination, following high accuracy with simply negative samples. This was the first DNNs model using the SDP, whose variants have been widely adopted in subsequent models.

Another way to improve semantic representation is the attention mechanism. Wang et al. [39] used two levels of attention mechanism, called Multi-Level Attention CNNs which enables end to end learning from task-specific labeled data. This multiple attention mechanism considered both the semantic information at the word level and sentence level. This model rarely uses any other external semantic information at all. At the first-level attention, the model constructed an entity-based attention matrix at the input level, which labeled the words related to the corresponding relation. At the second level, the mechanism captured more abstract high-level features to construct the final output matrix.

4.2 RNN- or LSTM-based methods

The general problems of CNNs in RE are that CNNs rarely consider global features and time sequence information, especially for the long-distance dependency between entity pairs. RNNs, LSMTs, and GRUs, the latest research methods for sequence modeling and problem transformation, can alleviate these problems. In this section, this paper will introduce some RNN-based methods, whose comparisons are shown in Table 5.

Structure-oriented: For learning relations within a long text and considering the timing information, Zhang et al. [40] used a bi-directional RNN architecture to this task. RNN combined the output of each hidden state and then represented the feature at the sentence level. At the end of the model, it conducted a max-pooling operation to pick up a few trigger word features for prediction. Although max-pooling operation simplified feature extraction, the effectiveness of these features remains to be discussed. Besides, the RNN model still has the problem of gradient explosion. To solve the problem of the gradient explosion, LSTM [31] was proposed by using the gate mechanism. Based on this, Xu et al. [46] proposed a model with LSTM (will be discussed in the semantic-oriented type).

With complete, sequential information about all words in the sentence is beneficial to RE, Zhang et al. [29] applied the bi-directional long short-term memory networks (BLSTM) to obtain the sentence level representation and also used several lexical features. The experiment results showed that using word embedding as input features alone was enough to achieve state-of-the-art results. This study documented the effectiveness of the BLSTM. Although the method improved the representation of sentence-level features, there were still two problems, a large number of introduced external artificial features and no effective feature filter mechanism.

Semantic oriented: Based on the above problems, Xu et al. [46] proposed a new DNNs model called SDP-LSTM. This model leveraged four types of information: Word vectors, POS tags, Grammatical relations, and WordNet hypernyms, to construct four channels for this model to support external information. And then, it concatenates the

Table 5 Comparison of RNN-based methods

Class	Author	Model name	Structure	Features set	Loss	Opt	F1
Str	Zhang et al. [40]	RNN	Bi-RNN +max-pooling	WE1+PI	C	SGD	80
				WE2+PI			82.5
	Zhang et al. [29]	BLSTM	BLSTM+max-pooling	WE3+PE+WN+NER+POS +WNSYN+Relative-DEP	–	–	84.3
				WE3+WA			82.7
Sem	Xu et al. [46]	SDP-LSTM	SDP+LSTM	WE2+SDP	C	SGD	82.4
				WE2+SDP+WA+POS+GR			83.7
	Zhou et al. [8]	Att-BLSTM	BLSTM+attention	WE3+PI	C	AdaDelta	84
		BLSTM	BLSTM+max-pooling	WE3+WA	–	–	82.7
	Xu et al. [63]	DRNNs	Multi channel RNN+max-pooling+augmentation	WE2+GR+POS+WN+ SDP(augmentation)	C	SGD	86.1
			Multi channel rnn+max-pooling	WE2+GR+POS+WN+SDP			84.16
	Xiao et al. [64]	BLSTM+BLSTM	2 level BLSTM+attention	WE2+WN+NER	R	AdaGrad	84.27
				WE2			83.9
	Qin et al. [41]	EAtt-BiGRU	Bi-GRU+entity attention	WE2+PE	–	AdaDelta	84.7
		–	Bi-GRU+random att				83.6
	Zhang et al. [42]	BiGUR-MCNN-ATT	BiGRU+multi size filter CNN attention	WE2+PE+SDP	–	AdaDelta	84.7
		BiGUR-Random-ATT	BiGRU+random attention				84.2
	Lee et al. [65]	BLSTM+LET	BLSTM+entity-aware attention+latent entity type	WE2+PE+LET	C	AdaDelta	85.2
		–	BLSTM+entity-aware attention	WE2+PE			84.7

Table symbols have the same meanings as Table 4. All of the F1 values are based on semeval2010-task8 [20]

result of the four channels to the softmax layer for prediction. This model is a little more complex than Zhang et al. [40] by considering a lot of additional syntaxes and semantic information.

Following the above work SDP-LSTM [46], to overcome the problem of shallow architecture that hardly represents the potential space in different network levels, Xu et al. [63] increased the neural network layers to tackle this challenge, with which this model captured the abstract features along the two sub-paths of SDP. Meanwhile, the small size of the semeval2010-task8 set and deeper neural networks may easily result in overfitting. Hence, the author augmented the data set by adding the directivity of the data based on the original SDP, avoiding the overfitting problem.

The SDP filters the input text but cannot filter the extracted features. To tackle this issue, Zhou et al. [8] came up with the attention mechanism in BLSTM, which automatically highlighted the important features only with the raw text instead of any other NLP toolkits or lexical resources. This work was a representative BLSTM model, and the architecture is shown in Fig. 7. Similar to the work of Zhou et al. [8], Xiao et al. [64] proposed a two-level BLSTM architecture with a two-level attention mechanism to extract a high-level representation of the raw sentence.

Although the attention mechanism gives more weight to the important features extracted by the model, this work [8] just presented a random weight, which lacked the consideration of prior knowledge. Therefore, the following works have improved this model.

Fed with entity pair and sentence, EAtt-BiGRU proposed by Qin et al. [41] leveraged the entity pair as prior knowledge to form attention weight. Different from Zhou et al.'s [8] work, EAtt-BiGRU applied bi-directional GRU (BiGRU) instead of BLSTM to reduce computation, which helped in obtaining the representation of sentence and adopted a one-way GRU to extract prior knowledge of entity pair. With the representation and prior knowledge, this model could generate the corresponding attention weight adaptively. This work improved the random

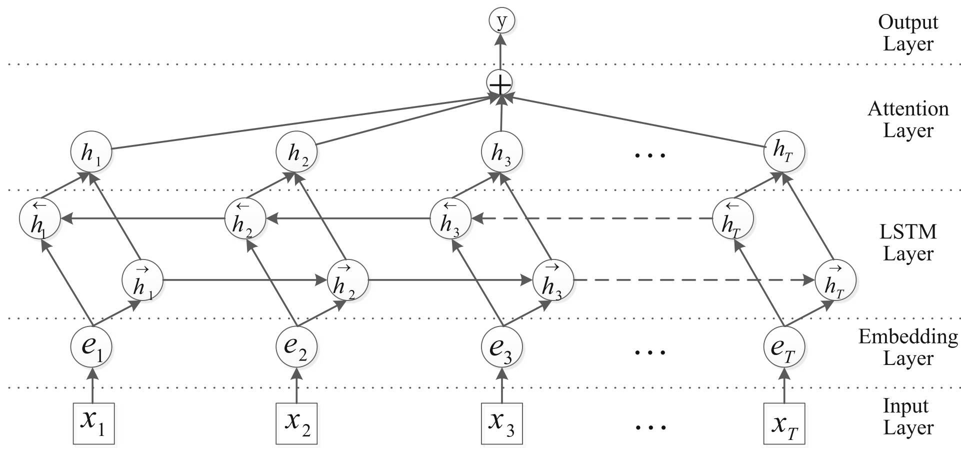

Fig. 7 Structure of the model [8], BLSTM with an attention mechanism

attention mechanism, but how to better integrate the prior knowledge needs further studying.

Zhang et al. [42] proposed another kind of attention mechanism based on the SDP, which was another prior knowledge. This model used Bi-GRU to extract sentence-level features and attention weights to select features from multi-channel CNN for classifying. Compared with other random or entity-based attention mechanisms [8, 41], this model constructed a better attention weight using the SDP.

Further research was followed by Lee et al. [65] who proposed a mixed model with BLSTM, self-attention, entity-aware attention, and latent entity typing module, getting state of the art without any high-level features. In general, the instance of the data set had no attribute of the entity type. However, the entity pair type was closely related to the relation classes. Previous works can only get word-level or sentence-level attention, but rarely obtain the degree of correlation between entities and other related words. Hence, this model introduced the latent entity typing module, self-attention module, entity-aware attention module, giving more prior knowledge.

4.3 Mix-structure-based methods

In addition to the above two types of models, some scholars combine these models based on their respective characteristics, which can be beneficial to RE task. There also exist two ways to merge these models: simply combination (**Structure-oriented**) [44] and attention mechanism (**Semantic-oriented**) [47]. The comparison of them is shown in Table 6.

Structure oriented: To integrate RNN and CNN, Zheng et al. [66] proposed two neural networks based on CNN and LSTM (MixCNN+CNN and MixCNN+LSTM) framework by joint learning the entity semantics and relation patterns. In this model, the entity semantic properties were reflected by their surrounding words, which could tackle the unknown words in entities (out-of-vocabulary problem), and the relation patterns modeled by the sub-sentence between the given entities instead of the whole sentence. With the entity semantics and relation patterns, the performance of RE could be improved. This research sheds new light on merging these two modules, showing the complementarity of the two modules as well as the necessity of module integration.

Zhang et al. [44] introduced the BLSTM-CNN, without any lexical attention mechanism or NLP toolkits. This model with three kinds of resources, word embedding, PE, and PI, showed that simply merging BLSTM and CNNs could perform better than any other single model. However, the PE and PI, showing the words around the nominals, are the same functions for RE which may result in overlap feeding.

Semantic oriented: Different from the above two works, Cai et al. [9] combined them depending on the SDP. To improve the model's sense of relation directivity, this model, called BRCNN, learned sentence features from the SDP in both positive and negative directions, which was beneficial for predicting the direction of the relation.

Following Cai et al.'s [9] work, Ren et al. [43] advanced a further work, a traditional CNN architecture with BRCNN [9], using two kinds of attention ('intra-cross') to combine the classification features coming from the original sentences and their corresponding descriptions. The description of an entity was from the external text, which can enrich the prior knowledge. Compared with different experiments and models, the result demonstrated that text descriptions can provide more features to model and replace WordNet to a certain extent in RE. In this line, this was the first RE method with entity description information. But the method of extracting description is too simple. As an external knowledge, description information should be closely related to the original sentence; therefore, the selection of description should be more targeted.

Guo et al. [47] proposed a novel Att-RCNN model to extract text features. This model leveraged GRU units

Table 6 Comparison of methods based on Mix neural networks

Class	Author	Model name	Structure	Features set	Loss	Opt	F1
Str	Zheng et al. [66]	MixCNN+CNN	Multi sizes filter CNN+fixed size filter CNN+max-pooling	WE2+WA	C	SGD	84.8
		MixCNN+LSTM	Multi sizes filter CNN+LSTM+max-pooling				83.8
	Zhang et al. [44]	BLSTM-CNN	BLSTM+fixed size filter CNN+max-pooling	WE2+PI+PE	–	–	83.2
		BLSTM-CNN+PF		WE2+PE			81.9
		BLSTM-CNN+PI		WE2+PI			82.1
Sem	Cai et al. [9]	BRCNN	2 level LSTM+2level fixed size filter CNN+SDP(2direction)	WE2+POS+NER+SDP	C	AdaDelta	86.3
				WE2+SDP			85.4
	Ren et al. [43]	DesRC(BRCNN)	BRCNN+fixed size filter CNN description+attention	WE2+PE+WN+SDP	–	SGD	87.4
		–	BRCNN+fixed size filter CNN	WE2+PE+SDP			84.7
	Guo et al. [47]	Att-RCNN	Fixed size filter CNN +max-pooling +BiGRU +attention	WE2+SDP	P	SGD	86.6
		–	Fixed size filter CNN+max-pooling+BiGRU				85.1
	Wang et al. [67]	Bi-SDP	Fixed size filter CNN +max-pooling +BLSTM +attention	WE2+PE+SDP	C	SGD	85.1

Table symbols have the same meanings as Table 4. In addition, the P refers to pairwise logistic loss. All of the F1 values are based on semeval2010-task8 [20]

instead of LSTM units, which have a higher speed in computing convergence and a more efficient CNNs to extract high-level features. Meanwhile, the two-level attention mechanism was similar to [39]. The special part of this model was the introduction of a new de-noise method, which can get a continuous fragment of the original text, based on the SDP [46].

Wang et al. [67] further utilized the SDP to construct a Bi-SDP with a parallel attention weight to cope with the direction of relation. In a sense, this method introduced more information about the SDP and interpreted another function (giving a hint of the direction of one relation) of a preposition in a sentence, which has been ignored by previous works.

4.4 Discussion

As stated above, the structure-oriented and semantic-oriented models improved the RE task in different aspects: one improved the ability of feature extraction and another focused on the semantic representation of text. From Tables 4, 5, and 6, we can find that semantic-oriented models exhibit more effective than structure-oriented models. This phenomenon has been widely observed in all of these models, which reflected that relational facts in a sentence were semantically strongly related to the sentence itself, and the model structure as well as related settings has a great impact on their performance. In other words, we argue that the constructed models around SDP, providing external semantic information, give an interpretable way of human-level thinking and cognition to cope with the RE task. Apart from the above works, some other research fields also have given new light on RE, such as distilling knowledge [68] and auxiliary learning [69]. Utilizing distilling knowledge method to generate soft labels and guiding the student network to learn latent knowledge could overcome the limitation of relational inventory or hard-label problem in supervised method. Auxiliary learning provided a simple module to further dig out the latent semantic relation in the wrong classified result, which alleviated the semantic gap between the sentence and the label. All of the above methods provided a solid foundation for the development of supervised RE. However, we have to admit that there are still many limitations in these approaches, some of which would be mitigated by distant supervision.

5 DNN-based Distant Supervision RE

The conception of distant supervision is introduced in Sect. 3.1. In the supervised method of RE, despite their excellent results, the insufficient training corpus is still an obstacle puzzling the development of RE. To solve this problem, Mintz et al. [58] proposed distant supervision, which is strongly based on an assumption that plays an important role in the selection of training instances. To date, this assumption has evolved into **Three assumptions**.

To solve the insufficient training samples problem, in 2009, Stanford University Professor Mintz et al. [58] proposed RE method of distant supervision at ACL conference. This method rarely needed manual annotation and generated large-scale training data sets (talked about that in Sect. 3.1). The large-scale training data sets are from these assumptions:

Assumption 1 If two entities participate in a relation, all sentences that mention these two entities express that relation.

Assumption 2 If two entities participate in a relation, at least one sentence that mentions these two entities might express that relation.

Assumption 3 A relation holding between two entities can be either expressed explicitly or inferred implicitly from all sentences that mention these two entities.

The first assumption [58] physically aligns the texts with a KB and trains a classifier heuristically using the existing triples in the KB. Riedel et al. [23] thought the first assumption was too strong resulting in the wrong label problem (noisy problem) and then proposed the second assumption, called: "one sentence from one bag." It leads to more accurate results. The third assumption [71] can consider more sentence features.

Based on the above three assumptions, distant supervision methods are composed of two research directions: **Sentence Encoder** optimizes the model and the performance of RE; **De-noise Algorithm** improves the quality of the data sets, which can be further subdivided into three main ways with different structures: the first makes full use of the word-level and sentence-level features of the instances in the bag for **enhancing representation**; the second introduces the **external knowledge**; while the third constructs a **plug-and-play component**. This section presents some related works and summarizes the evolution of this approach (shown in Fig. 8 (a)). The whole architecture of DNN-based distant supervision is shown in Fig. 8 (b), according to the above four points. Hence, the rest of this chapter will follow these four points: 1) encoder based, 2) representation based, 3) knowledge based, 4) plug and play based. In addiction, different from the supervised method, Table 7 does not include evaluation indicators, because most of these methods in Table 7 are just an approximate measure of precision.

5.1 Encoder-based methods

The encoder-based methods are a basic structure, including PCNN (Piecewise CNN), max-pooling, and multi-instance learning (MIL) modules, providing an infrastructure for distant supervision methods and have been referenced for improving subsequent de-noise methods.

PCNN+MIL: Inspired by Zeng et al. [36], Zeng et al. [70] exploited the PCNN with MIL, which divided the sentence into 3 segments based on the positions of two entities, to extract the relevant features automatically from a sentence and to get the important structural information. And then, they deployed MIL to train the model with the highest confidence level instance to reduce the noise (used one sentence from one bag). This model outperformed several competitive baselines but ignored other instances in the bag.

MIMLCNN: The same purpose as above, Jiang et al. [71] proposed a multi-instance multi-label CNN for distant supervision and introduced assumption 3. This work leveraged CNN to extract features from a single sentence and then aggregated all sentence representations into an entity pair level representation by cross-sentence max-pooling. In this way, the model merged features from different sentences, not one sentence from one bag. Meanwhile, the multi-relation pattern between entity pairs could be considered. Hence, for a given entity pair, the model predicted multiple relations simultaneously.

5.2 Representation-based methods

Unlike the previous simple encoder methods, Lin et al. [73, 74] introduced two novel attention models. The first used the weight of the attention mechanism to flag all instances in one bag, and then applied the vector weighted sum of all the sentences to represent a bag. Hence, this model could identify important instances from noisy sentences, as well as utilize all the information in the bag to optimize the performance. As a special case of the MIL, this model effectively reduced the influence of wrong label instances. The second one considered and leveraged multilingual corpus, which was based on [74]. This work proposed mono-lingual attention and a cross-lingual attention mechanism to excavate diverse information hidden in the data of different languages, and the experimental results showed that this work effectively modeled relation patterns among different languages. However, if each language builds a cross attention matrix, it does not seem realistic.

Yang et al. [72] and Du et al.'s [76] works are similar, both of them used two different kinds of attention mechanism (two-level) to extract features. The first one was used to present the sentence, which looks like [8], and alleviated the distractions of irrelevant words. The second one was like [74], leveraging attention mechanism to choose the weight of sentence with the highest probability.

In addition to using attention mechanisms, there are also ideas for incorporating more information to enhance the

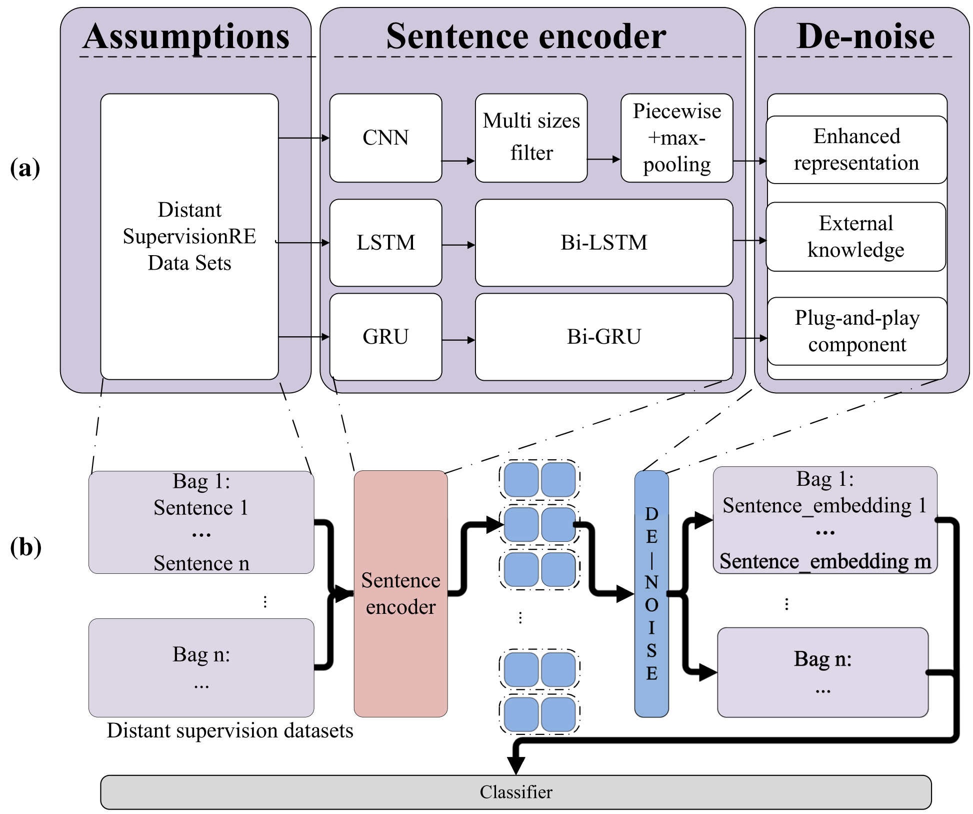

Fig. 8 (**a**) is the evolutionary process, and (**b**) is architecture of DNN-based distant supervision methods. (**b**) also shows the relationship between sentence encoder and de-noise method. The sentence encoder is responsible for encoding the sentence bag in the distant supervision data set into vectors, while the de-noise algorithm is responsible for selecting the sentences in each bag that can correctly represent the relationship

model. Banerjee et al. [75] proposed a simple co-occurrence based strategy method for calculating the highest confidence in distant supervision bag, which used the most frequent samples in the bag as the training label of the package. This was also a way to enhance the presentation.

5.3 Knowledge-based methods

Only with the entity pair and context, the extracted features are not enough to represent the sentence, which may result in poor performance. In recent years, with the development of word vectors and knowledge graphs, additional background information has been introduced in RE, improving the external description of relation vectors [77–79].

In Ji et al.'s [77] work, they continued to choose the classic PCNN model to get sentence feature vectors. But in the data source processing step, this work introduced the conception of relation vector from knowledge graph $r = e_1 - e_2$ to present the relation, and then, combined these two kinds of vectors to a new vector by concatenating. With the new vector, this model generated an attention weight vector to weight sum all of the sentence feature vectors as the bag's features. Meanwhile, to get additional background information, a description for entities from Freebase and Wikipedia pages was introduced to improve the entity representations. The experimental results showed that this method contained more background knowledge for entities.

Following the above work [77], Wang et al. [78] also introduced the conception of the relation vector of knowledge graph $r = e_1 - e_2$ to present the relation. But this model was different from the above, it leveraged the labeled data from the sentence itself, which meant that all the labels of the training data were determined by sentence and aligned entity pairs. In the end, the sentences, forming a sentence pattern, could be classified into a diverse group by the types of aligned entities in the knowledge graph. Hence, this method also alleviated the wrong label problem and made full use of the training corpus produced by distant supervision.

Since not all entities have description information, the above two methods may not apply to all entities, but some side information describing entities type or entity relations can be utilized. In Vashishth et al.'s [79] work, they considered some relevant and additional side information of KB, such as entity type and relation alias, and employed GCN to encode syntactic information from text. The entity type information, in this method, was integrated into an embedding to represent various entity types. And the relation alias came from some NLP toolkits (Stanford Open

Table 7 Comparison of DNN-based distant supervision methods

Class	Author	Model name	Structure	Features set	De-noise	Loss	Opt
En	Zeng et al. [70]	PCNN	Multi fixed size filter CNN+piecewise max-pooling	WE2+PE	Multi-instance Learning	C	Adadelta
	Jiang et al. [71]	MIMLCNN	Multi fixed size filter CNN+piecewise max-pooling+cross sentence max-pooling	WE2+PE	Output multi label	C	Adadelta
Re	Yang et al. [72]	BiGRU+2ATT	BiGRU+word level attention+sentence level attention	WE2+PE	Sentence level attention	C	Adam
	Lin et al. [73]	MNRE	Multi fixed size filter CNN+max-pooling+mono-lingual attention+cross-lingual attention	WE1+PE	mono-lingual (Sentence level attention)	–	SGD
	Lin et al. [74]	PCNN+ATT	Multi fixed size filter CNN+piecewise max-pooling+sentence level attention	WE2+PE	Selective attention	C	SGD
	Banerjee et al. [75]	MEM	Multi channel BLSTM	WE3+DP+POS	Co-occurrence statistics	C	–
	Du et al. [76]	MLSSA	BLSTM+word level attention+sentence level attention	WE2+PE	Sentence level attention	–	Adam
Ex	Ji et al. [77]	APCNNs+D	Multi fixed size filter CNN+piecewise max-pooling+fixed size filter CNN description+sentence level attention	WE2+PE	Sentence level attention	C	Adadelta
	Wang et al. [78]	LFDS	Multi fixed size filter CNN+piecewise max-pooling+word level attention	WE2+PE	KG Embedding	M	–
	Vashishth et al. [79]	RESIDE	GCN+BiGRU+word level attention+senten level attention	WE3+PE+DP	–	–	–
Pl	Qin et al. [80]	RL	Fixed size filter CNN+Reinforcement learning	WE2	Reinforcement Learning	–	–
	Qin et al. [81]	DSAGN	Fixed size filter CNN+GAN	WE2+PE	DSGAN	–	–

Table symbols have the same meanings as Table 4. In addition, the En, Re, Ex, Pl, M refer to sentence encoder, enhanced representation, external knowledge, plug-and-play component, margin loss, respectively

IE [82]) or Paraphrase database PPDB [83]. Both of these two kinds of description can also be seen as knowledge improving the performance of RE model.

5.4 Plug-and-play-based methods

Either the "enhanced representation" or "external knowledge" methods mentioned above are to improve RE model itself, which is not universal. Hence, to directly construct a method as a plug-and-play component in distant supervision to reduce the noise of data sets must be a new insight.

Qin et al. [80, 81] offered two approaches to constructing this plug-and-play component: One was the reinforcement learning framework for distant supervision [80] to solve false-positive case problems. The author argued that those wrong label sentences must be filtered by a hard decision, not by a soft weight of attention. In this way, this model generated less noise training data sets, which could be used in any previous state-of-the-art model. The second was DSGAN, using the idea of GAN to obtain a generator to classify the positive and negative samples in a bag from distant supervision. This is a kind of adversarial learning strategy, which could detect true-positive samples from the noisy distant supervision data sets. To some extent, these two methods alleviated the wrong label problem and formed a new high-confidence training data set. With this new data set, some recent state-of-the-art models have achieved further improvement in the experiment. Hence, both the "reinforcement learning framework" and DSGAN can be seen as a plug-and-play component in distant supervision. The structure of DSGAN is shown in Fig. 9.

5.5 Discussion

To summarize, distant supervision adopts various methods to get an excellent result. Fig. 8 and Table 7 illustrate that most of these works focus on de-noising algorithms. These works documented that although distant supervision brought substantial benefits, its drawbacks, the noise in the data set, were also obvious. The above three methods (representation based, knowledge based, and plug and play based) coped with the noise problem in different aspects, whereas they were still not optimal methods. Especially for knowledge-based methods, introducing external knowledge will enable models with the ability to commonsense reasoning, which is missing in RE task. Meanwhile, fewer models were devoted to feature extraction, which meant that the neural network variants have failed to significantly improve performance in pure text feature extraction. Consequently, purely improving the sentence encoder or de-noise algorithm only increases extra burden to the basic model instead of solving insufficient training corpus problem (We compare the characteristics of supervision and distant supervision methods in Table 8). And of course, some researchers are considering other solutions, which will be talked about later.

6 Data Sets

Variety of data sets exit for different methods; common data sets are SemEval 2010-task8 [20], ACE series 2003-2005, NYT+Freebase [23]. The SemEval 2010-task8 and ACE series are commonly used for supervised learning classification tasks, while the NYT+Freebase for distant supervision methods. Table 9 shows the comparison of 4 representative data sets. The following is a brief introduction to these data sets.

SemEval 2010-task8 was released in 2010, as an improvement on the Semeval 2007-task4; it provides a standard testbed for evaluating various methods of RE. This data set is used widely for evaluation, which contains 9 directional relations and an additional ’other’ relation, resulting in 19 relation classes in total. Most of the supervised methods discussed in this article use this data set. The relations in this data set are as follows:

- Cause-Effect: An event or object leads to an effect. (those cancers were caused by radiation exposures)
- Component-Whole: An object is a component of a larger whole. (my apartment has a large kitchen)
- Content-Container: An object is physically stored in a delineated area of space, the container. (Earth is located in the Milky Way)
- Entity-Destination: An entity is moving toward a destination. (the boy went to bed)
- Entity-Origin: An entity is coming or is derived from an origin, e.g., position or material. (letters from foreign countries)
- Message-Topic: An act of communication, written or spoken, is about a topic. (the lecture was about semantics)
- Member-Collection: A member forms a nonfunctional part of a collection. (there are many trees in the forest)
- Instrument-Agency: An agent uses an instrument. (phone operator)

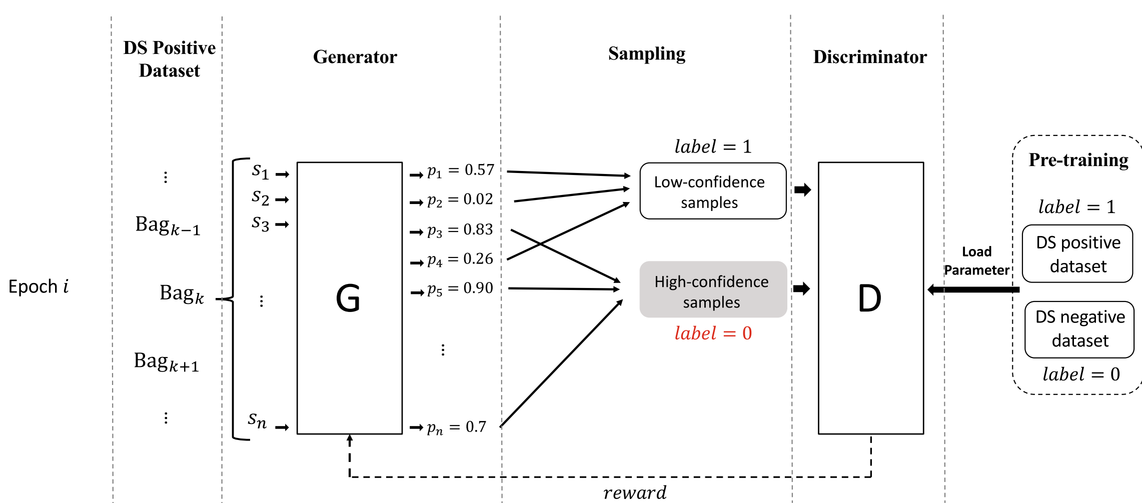

Fig. 9 Structure of model DSGAN [81]

Table 8 Comparison of supervised methods and distant supervision methods

Item	Features	Supervision	Distant Supervision
Data sets	Annotate mode	Manually annotated	Distant alignment KB
	Accuracy of labeled data	High accuracy	Low accuracy
	Noise	Low noise	High noise
	Data Size	Small	Large
Applicability	Model portability	Low	High
	Cross-domain applicability	Low	High
Accuracy of prediction	–	High	Low

Table 9 Comparison of data sets

Data set	Relations	Data Amount (words/language)
SemEval-2010 Task8	18	10717
ACE04	24	350K
NYT+Freebase	53	695059
FewRel	100	70000

- Product-Agency: A producer causes a product to exist. (a factory manufactures suits)
- Other: If none of the above nine relations appear to be suitable.

This data set contains 10717 labeled data, of which 8000 are used for training and the remaining 2717 for testing. The nominals of the sentences in the data set are marked by XML tags. Table 10 shows an example of labeled sentences.

ACE2003-2005 Series comes from LDC (Linguistic Data consortium), consisting of various types of annotated for entities and relations. In three different years of the corpus (ACE 2003, 2004, 2005), the ACE tasks are more complex than SemEval 2010-task8. The corpus can be divided into broadcast news, newswire, and telephone conversations, including the complete set of English, Arabic, Chinese training data. So the annotated entities in this corpus are pronoun words or other irregular words. In addition to the task of RE, they can also be applied in the following five tasks: Entity Detection and Recognition, Entity Mention Detection, EDR Co-reference, Relation Mention Detection, and Relation Detection and Recognition of given reference entities. This data set is typically used in supervised models. Table 11 shows the introduction of ACE data sets.

NYT+Freebase means New York times + Freebase, which is a common way to generate data sets in distant supervision. The distant supervision method produces data sets by extracting relations through the heuristic alignment of text and entities in the KB (Section 3.2). As a result, in this way, distant supervision creates large-scale training data automatically. This process is shown in Fig. 4. The most widely used data set is generated by Riedel et al. [23], which has 522611 training sentences and 172448 test sentences, labeled by 53 candidate relations in Freebase, and an extra-label of NA (nearly 80% of the sentences in the training data are labeled as NA).

Other Data sets include SemEval2017-task10 [84], SemEval2018-task7 [85], TACRED [86], KBP37 [40], DDIExtraction2011 [87], DDIExtraction2013 [88], FewRel [21], FewRel2 [22].

7 Challenges of RE

Based on the above elaboration, both supervised and distant supervision methods have their own characteristics. Supervised methods are better suited for specific domain, while distant supervision methods are better for generic domains. As a result, it is difficult to specify which methods are currently the best. Hence, we compare the characteristics of supervised and distant supervision

Table 10 Some examples from SemEval 2010-task8

Example1:	<e1>People</e1> have been moving back into <e2>downtown</e2>.
Relation:	Entity-Destination(e1,e2).
Example2:	Cieply's <e1>story</e1> makes a compelling <e2>point</e2> about modern-day studio economics.
Relation:	Message-Topic(e1,e2).

Table 11 An introduction to ACE data sets

Corpus	Data Amount (words/language)	Tasks	Languages
ACE03	100 K training 50 K evaluation	Entities Relations	Chinese English Arabic
ACE04	300 K training 50 K evaluation	Entities Relations Events	Chinese English Arabic
ACE05	750 K training 150 K evaluation	Entities Relations Events	Chinese English Arabic

methods in Table 8. Nevertheless, although RE task has made great progress, this field still faces many challenges. We argue that these challenges could be divided into two categories based on the current situation: **theories** and **scenarios**.

7.1 Challenges from theories

All the works mentioned in this article have solved three theoretical problems: **enhanced representation** for semantics, **structure exploration** for DNNs, and **de-noise method** for text, having important guiding significance for future research; especially, with the development of pre-training models, such as BERT [89] and RoBERT [90], not included in our work, further greatly improved the semantic representation, and have triggered a new wave of research. Meanwhile, structure exploration and de-noise methods, such as attention mechanism, SDP, and even feature set or loss function in our work, could also be referenced in future BERT based work. In addition to these problems, the following challenges also need focusing.

Transfer Learning: is a cross-domain adaptive solution. It provides better domain transfer capability for RE model, especially in supervised learning methods, making the model have better extensibility to another corpus. Presently, world knowledge and the number of relations are not stagnant, which need a dynamic KB to contain this knowledge. Nevertheless, most RE models are explored in predefined relation inventory with low utilization of dynamic knowledge and insufficient data sets. To keep the model continuously receiving new training samples and making full use of other related corpus is worthy of exploration. It is noted that the results in this field are not yet significant [91–93].

Relationship Framework: limits RE task in an established framework, which may result in a semantic relation out of the predefined relation inventory (out-of-vocabulary problem). Different data sets have different definitions of relation inventories, which makes the models trained based on data sets have limited adaptability to different fields. If the industry can construct a framework with a unified description of all kinds of relations and make RE have a hierarchical structure, then it may have a better multi-domain adaptation. Predecessors have done a lot of work [20] on the definition of relation inventories, but no agreement has been reached.

Overlapped Multiple Relations: is a confusing problem, because most RE methods extract relation from one entity pair. But in fact, one entity pair may have multiple relations in a sentence, which still keeps challenging. From the point of the input data, it is difficult to give multiple outputs from the same input. In other words, one entity pair expresses multiple relations, confusing a model seriously. To address this issue, Zhang et al. [94] came up with a RE approach based on capsule networks with an attention mechanism, which outputs multi-relations of one entity pair. This study was the first attempt that applied capsule network to solve this challenge. To some extent, this research is an important step.

Relationship Reasoning: presents a way to synthesize background knowledge and be aware of new knowledge based on logical ability. For RE, with the reasoning ability and some existing relational facts, we may use this logical ability to expand our existing knowledge. At the same time, in the open domain, without the relation inventory in advance, it is a very forward-looking work to obtain the relation directly from the existing corpus in the real world. Hence, using background knowledge to obtain the relation by reasoning is an exciting way to solve the problem. Several attempts [43, 77, 78] have been made to use a knowledge graph or introduce entity description information to RE. However, how to further improve the reasoning mechanism still needs studying.

Others: In addition to these theoretical challenges, several problems in the existing methods also need solving. One is the problem of error propagation in supervised methods, and the other is the problem of wrong label in distant supervision. Especially for the latter one, if the wrong label problem can be well solved, a large amount of effective sample data will be obtained, which will make an important contribution to this field.

7.2 Challenges from scenarios

In addition to the above theoretical challenges, RE task also faces the challenges from different scenarios due to the different presentation of data sets. Generally speaking, they are as follows:

Cross-sentence RE: or document-level RE is likewise one significant research field. The current RE task mainly focuses on processing the entity pair relation in intra-sentence. In the actual situation, most entities in the text represent one relation by multiple sentences. Current models may not be able to handle such tasks directly. In particular, distant supervision generates a large scale of document-level corpora, which can only be solved by a cross-sentence RE technique. Some researchers proposed a series of novel methods [95–99] to obtain entity relation across sentences. These studies offered some important insights into cross-sentence RE.

Joint Extraction: integrates name entity recognizer and RE into one task, an end-to-end job, yielding the relation and entity pair together. In general, these methods [100–104] reduces the possibility of error propagation, compared with pure RE task, they have a good prospect, but their practicality need further improving.

Few-shot RE: is another challenge for RE. As stated above, the insufficient training instance puzzle RE task. In some past experiments, we can observe that when the number of a certain type of instance in the data set is small, the corresponding model classification effect is poor. Few-shot scenario resolves this obstacle in a new form, which utilizes a few instances to train a model, and then classify entity pairs. Han et al. [21] and Gao et al. [22] provided two large-scale supervised Few-shot relation extraction data sets to promote this task. Nevertheless, any text data are noisy and its relation pattern is complex. If a model only depends on a small sample and outputs a good performance, the model must have a very accurate concept of a certain relation pattern. Therefore, this scenario puts forward very high requirements for RE task.

8 Conclusions

This article begins by laying out the general framework and some basic concepts, survey on DNN-based methods in supervised and distant supervision, and further discuss some promising directions. Different from previous surveys, we mainly focus on how various model structures evolution and settings affect the RE task, which has rarely been investigated in previous research. Hence, we extensively survey the previous RE literature on variant models to show how they effect model development, and progress in the current RE field. To facilitate reader's understanding, supervised and distant supervision approaches are classified into 2 and 4 categories, respectively. Then, we make sufficient comparison from the perspective of the model structures and settings. Through our paper, we hope to present a clear process of how the current work promotes the RE task and hope to attract more researchers' attention.

Acknowledgements This work was supported by National Natural Science Foundation of China (No. U19A2059), the Fundamental Research Funds for the Central Universities (NO. ZYGX2020ZB034), and Sichuan Science and Technology Program (NO. 2019YFG0507 & 2020YFG0328). We sincerely thank Mr. Kombou Victor, Anto Leoba Jonathan and Wilson Jim Owusu for their helpful discussions.

Declaration

Conflicts of interest The authors declare that they have no conflict of interest regarding publication of this paper.

References

1. Bollacker K, Evans C, Paritosh P, Sturge T, Taylor J (2008) Freebase: a collaboratively created graph database for structuring human knowledge. In: Proceedings of the 2008 ACM SIGMOD international conference on Management of data, 1247–1250
2. Auer S, Bizer C, Kobilarov G, Lehmann J, Cyganiak R, Ives Z (2007) Dbpedia: a nucleus for a web of open data. In: The semantic web, Springer, 722–735
3. Vrandečić D, Krötzsch M (2014) Wikidata: a free collaborative knowledgebase. Commun ACM 57(10):78–85
4. Suchanek FM, Kasneci G, Weikum G (2007) Yago: a core of semantic knowledge. In: Proceedings of the 16th international conference on World Wide Web, 697–706
5. Brin S (1998) Extracting patterns and relations from the world wide web. In: International workshop on the world wide web and databases, Springer, 172–183
6. Agichtein E, Gravano L (2000) Snowball: Extracting relations from large plain-text collections. In: Proceedings of the fifth ACM conference on digital libraries, ACM, 85–94
7. Liu C, Sun W, Chao W, Che W (2013) Convolution neural network for relation extraction. In: International conference on advanced data mining and applications, Springer, 231–242
8. Zhou P, Shi W, Tian J, Qi Z, Li B, Hao H, Xu B (2016) Attention-based bidirectional long short-term memory networks for relation classification. In: Proceedings of the 54th annual meeting of the association for computational linguistics (Volume 2: Short Papers), vol 2, 207–212
9. Cai R, Zhang X, Wang H (2016) Bidirectional recurrent convolutional neural network for relation classification. In: Proceedings of the 54th annual meeting of the association for computational linguistics (Volume 1: Long Papers), vol 1, 756–765
10. Nédellec C, Bossy R, Kim JD, Kim JJ, Ohta T, Pyysalo S, Zweigenbaum P (2013) Overview of bionlp shared task 2013. In: Proceedings of the BioNLP shared task 2013 workshop, 1–7
11. Vazquez M, Krallinger M, Leitner F, Valencia A (2011) Text mining for drugs and chemical compounds: methods, tools and applications. Mol Inf 30(6–7):506–519
12. Vela M, Declerck T (2009) Concept and relation extraction in the finance domain. In: Proceedings of the eight international conference on computational semantics, 346–350
13. Kumar S (2017) A survey of deep learning methods for relation extraction. arXiv preprint arXiv:170503645
14. Pawar S, Palshikar GK, Bhattacharyya P (2017) Relation extraction: a survey. arXiv preprint arXiv:171205191
15. Smirnova A, Cudré-Mauroux P (2018) Relation extraction using distant supervision: A survey. ACM Computing Surveys (CSUR) 51(5):1–35

16. Nayak T, Majumder N, Goyal P, Poria S (2021) Deep neural approaches to relation triplets extraction: a comprehensive survey. arXiv preprint arXiv:210316929
17. Aydar M, Bozal O, Ozbay F (2020) Neural relation extraction: a survey. arXiv e-prints pp arXiv–2007
18. Han X, Gao T, Lin Y, Peng H, Yang Y, Xiao C, Liu Z, Li P, Sun M, Zhou J (2020) More data, more relations, more context and more openness: a review and outlook for relation extraction. arXiv preprint arXiv:200403186
19. Liu K (2020) A survey on neural relation extraction. Science China Technological Sciences pp 1–19
20. Hendrickx I, Kim SN, Kozareva Z, Nakov P, Ó Séaghdha D, Padó S, Pennacchiotti M, Romano L, Szpakowicz S (2009) Semeval-2010 task 8: multi-way classification of semantic relations between pairs of nominals. In: proceedings of the workshop on semantic evaluations: recent achievements and future directions, Association for computational linguistics, 94–99
21. Han X, Zhu H, Yu P, Wang Z, Yao Y, Liu Z, Sun M (2018) Fewrel: A large-scale supervised few-shot relation classification dataset with state-of-the-art evaluation. arXiv preprint arXiv:181010147
22. Gao T, Han X, Zhu H, Liu Z, Li P, Sun M, Zhou J (2019) Fewrel 2.0: towards more challenging few-shot relation classification. arXiv preprint arXiv:191007124
23. Riedel S, Yao L, McCallum A (2010) Modeling relations and their mentions without labeled text. Joint European conference on machine learning and knowledge discovery in databases. Springer, Heidelberg, pp 148–163
24. Tang W, Hui B, Tian L, Luo G, He Z, Cai Z (2021) Learning disentangled user representation with multi-view information fusion on social networks. Inf Fusion 74:77–86
25. LeCun Y, Boser B, Denker JS, Henderson D, Howard RE, Hubbard W, Jackel LD (1989) Backpropagation applied to handwritten zip code recognition. Neural Comput 1(4):541–551
26. Elman JL (1991) Distributed representations, simple recurrent networks, and grammatical structure. Mach Learn 7(2–3):195–225
27. Socher R, Huval B, Manning CD, Ng AY (2012) Semantic compositionality through recursive matrix-vector spaces. In: Proceedings of the 2012 joint conference on empirical methods in natural language processing and computational natural language learning, Association for computational linguistics, 1201–1211
28. Scarselli F, Gori M, Tsoi AC, Hagenbuchner M, Monfardini G (2008) The graph neural network model. IEEE Trans Neural Networks 20(1):61–80
29. Zhang S, Zheng D, Hu X, Yang M (2015) Bidirectional long short-term memory networks for relation classification. In: Proceedings of the 29th Pacific Asia conference on language, information and computation, 73–78
30. Sundermeyer M, Schlüter R, Ney H (2012) Lstm neural networks for language modeling. In: Thirteenth annual conference of the international speech communication association
31. Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9(8):1735–1780
32. Chung J, Gulcehre C, Cho K, Bengio Y (2014) Empirical evaluation of gated recurrent neural networks on sequence modeling. arXiv preprint arXiv:14123555
33. Mikolov T, Chen K, Corrado G, Dean J (2013) Efficient estimation of word representations in vector space. arXiv preprint arXiv:13013781
34. Turian J, Ratinov L, Bengio Y (2010) Word representations: a simple and general method for semi-supervised learning. In: Proceedings of the 48th annual meeting of the association for computational linguistics, Association for computational linguistics, 384–394
35. Pennington J, Socher R, Manning C (2014) Glove: Global vectors for word representation. In: Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP), 1532–1543
36. Zeng D, Liu K, Lai S, Zhou G, Zhao J, et al. (2014) Relation classification via convolutional deep neural network
37. Nguyen TH, Grishman R (2015) Relation extraction: perspective from convolutional neural networks. In: Proceedings of the 1st workshop on vector space modeling for natural language processing, 39–48
38. Santos CNd, Xiang B, Zhou B (2015) Classifying relations by ranking with convolutional neural networks. arXiv preprint arXiv:150406580
39. Wang L, Cao Z, De Melo G, Liu Z (2016) Relation classification via multi-level attention cnns. In: Proceedings of the 54th annual meeting of the association for computational linguistics (volume 1: long papers), 1298–1307
40. Zhang D, Wang D (2015) Relation classification via recurrent neural network. arXiv preprint arXiv:150801006
41. Qin P, Xu W, Guo J (2017) Designing an adaptive attention mechanism for relation classification. In: 2017 International joint conference on neural networks (IJCNN), IEEE, 4356–4362
42. Zhang C, Cui C, Gao S, Nie X, Xu W, Yang L, Xi X, Yin Y (2019) Multi-gram cnn-based self-attention model for relation classification. IEEE Access 7:5343–5357
43. Ren F, Zhou D, Liu Z, Li Y, Zhao R, Liu Y, Liang X (2018) Neural relation classification with text descriptions. In: Proceedings of the 27th international conference on computational linguistics, 1167–1177
44. Zhang L, Xiang F (2018) Relation classification via bilstm-cnn. In: International conference on data mining and big data, Springer, pp 373–382
45. Mooney RJ, Bunescu RC (2006) Subsequence kernels for relation extraction. In: Advances in neural information processing systems, pp 171–178
46. Xu Y, Mou L, Li G, Chen Y, Peng H, Jin Z (2015) Classifying relations via long short term memory networks along shortest dependency paths. In: proceedings of the 2015 conference on empirical methods in natural language processing, 1785–1794
47. Guo X, Zhang H, Yang H, Xu L, Ye Z (2019) A single attention-based combination of cnn and rnn for relation classification. IEEE Access 7:12467–12475
48. Jin L, Song L, Zhang Y, Xu K, Ma Wy YuD (2020) Relation extraction exploiting full dependency forests. Proc AAAI Conf Artif Intell 34:8034–8041
49. Hearst MA (1992) Automatic acquisition of hyponyms from large text corpora. In: Proceedings of the 14th conference on Computational linguistics-Volume 2, Association for computational linguistics, pp 539–545
50. Berland M, Charniak E (1999) Finding parts in very large corpora. In: Proceedings of the 37th annual meeting of the association for computational linguistics
51. Etzioni O, Cafarella M, Downey D, Kok S, Popescu AM, Shaked T, Soderland S, Weld DS, Yates A (2004) Web-scale information extraction in knowitall:(preliminary results). In: Proceedings of the 13th international conference on World Wide Web, ACM, 100–110
52. Yates A, Cafarella M, Banko M, Etzioni O, Broadhead M, Soderland S (2007) Textrunner: open information extraction on the web. In: Proceedings of Human Language Technologies: The Annual Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations, Association for Computational Linguistics, pp 25–26

53. Phi VT, Santoso J, Shimbo M, Matsumoto Y (2018) Ranking-based automatic seed selection and noise reduction for weakly supervised relation extraction. In: Proceedings of the 56th Annual Meeting of the association for computational linguistics (Volume 2: Short Papers), pp 89–95
54. Hasegawa T, Sekine S, Grishman R (2004) Discovering relations among named entities from large corpora. In: Proceedings of the 42nd annual meeting on association for computational linguistics, association for computational linguistics, p 415
55. Rink B, Harabagiu S (2010) Utd: Classifying semantic relations by combining lexical and semantic resources. In: Proceedings of the 5th International workshop on semantic evaluation, Association for computational linguistics, 256–259
56. Kambhatla N (2004) Combining lexical, syntactic, and semantic features with maximum entropy models for extracting relations. In: Proceedings of the ACL 2004 on interactive poster and demonstration sessions, Association for computational linguistics, p 22
57. Bunescu RC, Mooney RJ (2005) A shortest path dependency kernel for relation extraction. In: Proceedings of the conference on human language technology and empirical methods in natural language processing, Association for computational linguistics, 724–731
58. Mintz M, Bills S, Snow R, Jurafsky D (2009) Distant supervision for relation extraction without labeled data. In: Proceedings of the joint conference of the 47th annual meeting of the ACL and the 4th international joint conference on natural language processing of the AFNLP: Volume 2-Volume 2, Association for computational linguistics, 1003–1011
59. Manning C, Surdeanu M, Bauer J, Finkel J, Bethard S, McClosky D (2014) The stanford corenlp natural language processing toolkit. In: Proceedings of 52nd annual meeting of the association for computational linguistics: system demonstrations, 55–60
60. Xu K, Feng Y, Huang S, Zhao D (2015) Semantic relation classification via convolutional neural networks with simple negative sampling. arXiv preprint arXiv:150607650
61. Kim Y (2014) Convolutional neural networks for sentence classification. arXiv preprint arXiv:14085882
62. Kalchbrenner N, Grefenstette E, Blunsom P (2014) A convolutional neural network for modelling sentences. arXiv preprint arXiv:14042188
63. Xu Y, Jia R, Mou L, Li G, Chen Y, Lu Y, Jin Z (2016) Improved relation classification by deep recurrent neural networks with data augmentation. arXiv preprint arXiv:160103651
64. Xiao M, Liu C (2016) Semantic relation classification via hierarchical recurrent neural network with attention. In: Proceedings of COLING 2016, the 26th international conference on computational linguistics: Technical Papers, 1254–1263
65. Lee J, Seo S, Choi YS (2019) Semantic relation classification via bidirectional lstm networks with entity-aware attention using latent entity typing. arXiv preprint arXiv:190108163
66. Zheng S, Xu J, Zhou P, Bao H, Qi Z, Xu B (2016) A neural network framework for relation extraction: Learning entity semantic and relation pattern. Knowl-Based Syst 114:12–23
67. Wang H, Qin K, Lu G, Luo G, Liu G (2020) Direction-sensitive relation extraction using bi-sdp attention model. Knowl-Based Syst 21:105928105928
68. Zhang Z, Shu X, Yu B, Liu T, Zhao J, Li Q, Guo L (2020) Distilling knowledge from well-informed soft labels for neural relation extraction. In: AAAI, pp 9620–9627
69. Lyu S, Cheng J, Wu X, Cui L, Chen H, Miao C (2020) Auxiliary learning for relation extraction. IEEE Trans Emer Top Comp Intell
70. Zeng D, Liu K, Chen Y, Zhao J (2015) Distant supervision for relation extraction via piecewise convolutional neural networks. In: Proceedings of the 2015 conference on empirical methods in natural language processing, 1753–1762
71. Jiang X, Wang Q, Li P, Wang B (2016) Relation extraction with multi-instance multi-label convolutional neural networks. In: Proceedings of COLING 2016, the 26th international conference on computational linguistics: Technical papers, pp 1471–1480
72. Yang L, Ng TLJ, Mooney C, Dong R (2017) Multi-level attention-based neural networks for distant supervised relation extraction. In: AICS, pp 206–218
73. Lin Y, Liu Z, Sun M (2017) Neural relation extraction with multi-lingual attention. In: Proceedings of the 55th annual meeting of the association for computational linguistics (Volume 1: Long Papers), pp 34–43
74. Lin Y, Shen S, Liu Z, Luan H, Sun M (2016) Neural relation extraction with selective attention over instances. In: Proceedings of the 54th annual meeting of the association for computational linguistics (Volume 1: long papers), vol 1, 2124–2133
75. Banerjee S, Tsioutsiouliklis K (2018) Relation extraction using multi-encoder lstm network on a distant supervised dataset. In: 2018 IEEE 12th international conference on semantic computing (ICSC), IEEE, pp 235–238
76. Du J, Han J, Way A, Wan D (2018) Multi-level structured self-attentions for distantly supervised relation extraction. arXiv preprint arXiv:180900699
77. Ji G, Liu K, He S, Zhao J (2017) Distant supervision for relation extraction with sentence-level attention and entity descriptions. In: Thirty-first AAAI conference on artificial intelligence
78. Wang G, Zhang W, Wang R, Zhou Y, Chen X, Zhang W, Zhu H, Chen H (2018) Label-free distant supervision for relation extraction via knowledge graph embedding. In: Proceedings of the 2018 conference on empirical methods in natural language processing, 2246–2255
79. Vashishth S, Joshi R, Prayaga SS, Bhattacharyya C, Talukdar P (2018) Reside: Improving distantly-supervised neural relation extraction using side information. In: Proceedings of the 2018 conference on empirical methods in natural language processing, 1257–1266
80. Qin P, Xu W, Wang WY (2018a) Robust distant supervision relation extraction via deep reinforcement learning. arXiv preprint arXiv:180509927
81. Qin P, Xu W, Wang WY (2018b) Dsgan: Generative adversarial training for distant supervision relation extraction. arXiv preprint arXiv:180509929
82. Angeli G, Premkumar MJJ, Manning CD (2015) Leveraging linguistic structure for open domain information extraction. In: Proceedings of the 53rd annual meeting of the association for computational linguistics and the 7th international joint conference on natural language processing (Volume 1: Long Papers), vol 1, pp 344–354
83. Pavlick E, Rastogi P, Ganitkevitch J, Van Durme B, Callison-Burch C (2015) Ppdb 2.0: Better paraphrase ranking, fine-grained entailment relations, word embeddings, and style classification. In: proceedings of the 53rd annual meeting of the association for computational linguistics and the 7th international joint conference on natural language processing (Volume 2: Short Papers), vol 2, pp 425–430
84. Bethard S, Carpuat M, Cer D, Jurgens D, Nakov P, Zesch T (2016) Proceedings of the 10th international workshop on semantic evaluation (semeval-2016). In: Proceedings of the 10th international workshop on semantic evaluation (SemEval-2016)
85. Gábor K, Buscaldi D, Schumann AK, QasemiZadeh B, Zargayouna H, Charnois T (2018) Semeval-2018 task 7: Semantic relation extraction and classification in scientific papers. In: Proceedings of the 12th international workshop on semantic evaluation, pp 679–688

86. Zhang Y, Zhong V, Chen D, Angeli G, Manning CD (2017) Position-aware attention and supervised data improve slot filling. In: Proceedings of the 2017 conference on empirical methods in natural language processing (EMNLP 2017), 35–45, https://nlp.stanford.edu/pubs/zhang2017tacred.pdf
87. Segura Bedmar I, Martinez P, Sánchez Cisneros D (2011) The 1st ddiextraction-2011 challenge task: Extraction of drug-drug interactions from biomedical texts
88. Segura Bedmar I, Martínez P, Herrero Zazo M (2013) Semeval-2013 task 9: Extraction of drug-drug interactions from biomedical texts (ddiextraction 2013). Association for computational linguistics
89. Devlin J, Chang MW, Lee K, Toutanova K (2018) Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:181004805
90. Liu Y, Ott M, Goyal N, Du J, Joshi M, Chen D, Levy O, Lewis M, Zettlemoyer L, Stoyanov V (2019) Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:190711692
91. Di S, Shen Y, Chen L (2019) Relation extraction via domain-aware transfer learning. In: Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & Data Mining, 1348–1357
92. Sun C, Wu Y (2019) Distantly supervised entity relation extraction with adapted manual annotations. Proc AAAI Conf Artif Intell 33:7039–7046
93. Zhang N, Deng S, Sun Z, Chen J, Zhang W, Chen H (2019) Transfer learning for relation extraction via relation-gated adversarial learning. arXiv preprint arXiv:190808507
94. Zhang N, Deng S, Sun Z, Chen X, Zhang W, Chen H (2018) Attention-based capsule networks with dynamic routing for relation extraction. arXiv preprint arXiv:181211321
95. Sahu SK, Christopoulou F, Miwa M, Ananiadou S (2019) Inter-sentence relation extraction with document-level graph convolutional neural network. arXiv preprint arXiv:190604684
96. Guo Z, Zhang Y, Lu W (2019) Attention guided graph convolutional networks for relation extraction pp 241–251, 10.18653/v1/p19-1024, arxiv:1906.07510
97. Zhang Y, Qi P, Manning CD (2019) Graph convolution over pruned dependency trees improves relation extraction (2005):2205–2215, 10.18653/v1/d18-1244, arxiv:1809.10185
98. Wang H, Qin K, Lu G, Yin J, Zakari RY, Owusu JW (2021) Document-level relation extraction using evidence reasoning on rst-graph. Knowledge-based systems p 107274
99. Zhang N, Chen X, Xie X, Deng S, Tan C, Chen M, Huang F, Si L, Chen H (2021) Document-level relation extraction as semantic segmentation. arXiv preprint arXiv:210603618
100. Ye H, Chao W, Luo Z, Li Z (2016) Jointly extracting relations with class ties via effective deep ranking. arXiv preprint arXiv:161207602
101. Miwa M, Bansal M (2016) End-to-end relation extraction using lstms on sequences and tree structures. arXiv preprint arXiv:160100770
102. Li F, Zhang M, Fu G, Ji D (2017) A neural joint model for entity and relation extraction from biomedical text. BMC Bioinform 18(1):198
103. Zheng S, Wang F, Bao H, Hao Y, Zhou P, Xu B (2017) Joint extraction of entities and relations based on a novel tagging scheme. arXiv preprint arXiv:170605075
104. Xiao Y, Tan C, Fan Z, Xu Q, Zhu W (2020) Joint entity and relation extraction with a hybrid transformer and reinforcement learning based model. In: AAAI, pp 9314–9321